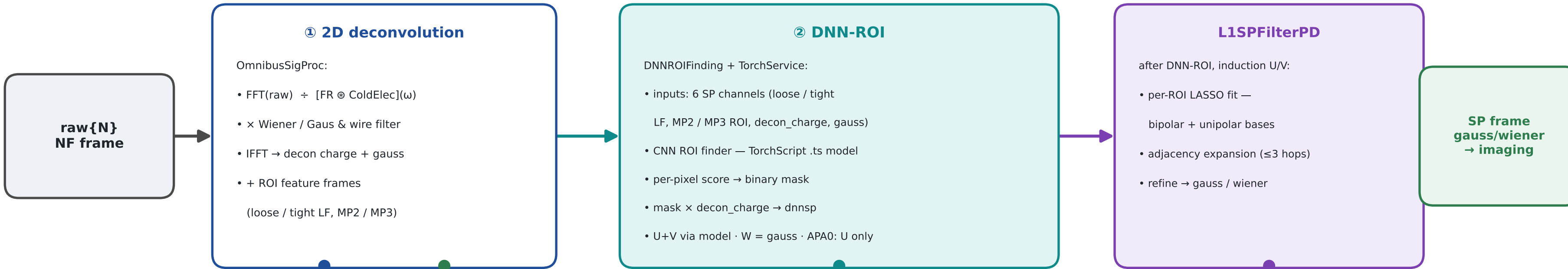
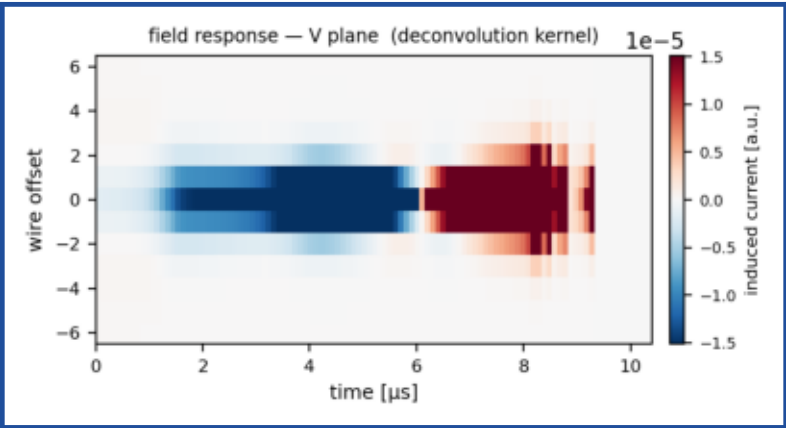


ProtoDUNE-HD Wire-Cell Signal Processing · OmnibusSigProc → DNN-ROI → L1SPFilterPD (per APA)

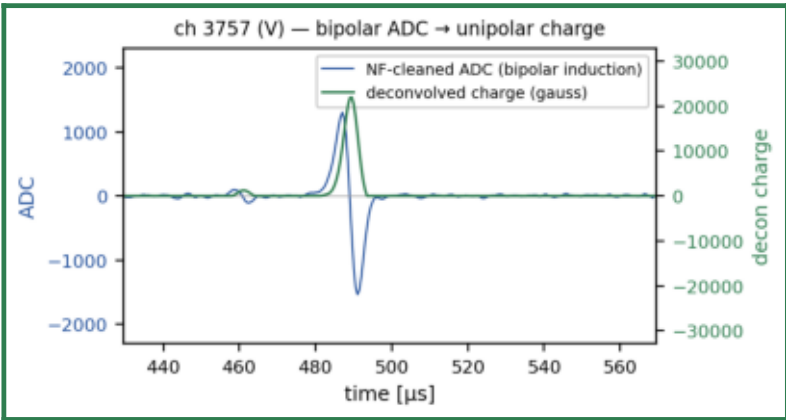
$$\text{deconvolve } \frac{\text{FFT}[\text{raw}] \cdot G_{\text{filter}}(\omega)}{[\text{FR} * E](\omega)} \rightarrow \text{neural-network ROI} \rightarrow \text{L1SP unipolar fit}$$



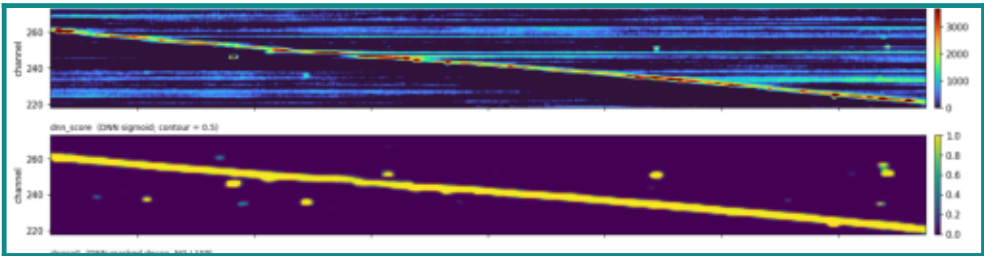
W plane: standard SP gauss (no DNN). APA0 specials: own field-response file · U↔V plane2layer swap · DNN-ROI on U only (V anomalous).



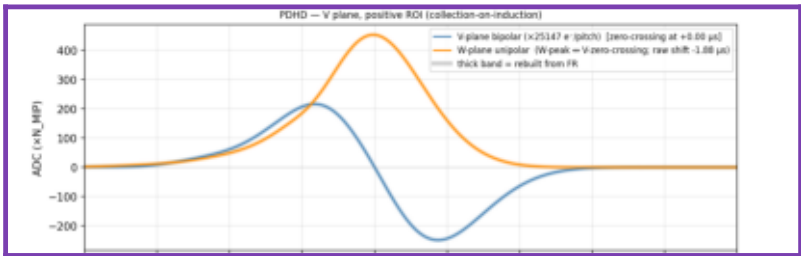
field-response kernel (V)



raw ADC → deconvolved charge (data)



DNN-ROI: deconvolved charge → CNN score (data)



L1SP response bases (V bipolar / W unipolar)